15 JANUARY 2026

LIVERPOOL HOPE UNIVERSITY
1844

DATA ANALYTICS

ONLINE SHOPPING BEHAVIOUR ANALYSIS

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1. Introduction:

Online shopping has become an important part of modern consumer behaviour, with more people using digital platforms to buy many different products. The growth of e-commerce has changed traditional shopping habits by offering greater convenience, a wider choice of products, and faster delivery options. As online shopping continues to grow in popularity, it is important to understand the factors that affect how often consumers use online platforms. Studying these behaviour patterns provides useful insight into consumer preferences and the wider business environment.

The dataset used in this study includes 95 survey responses that record demographic details, shopping behaviour, and views on key online shopping factors. These include age group, gender, shopping frequency, shopping preference, and the perceived importance of factors such as delivery, security, convenience, product range, and website quality. Using both demographic and opinion-based variables makes it possible to explore links between consumer characteristics and online shopping habits.

This report aims to examine these variables using a clear set of statistical methods that are suitable for categorical data and Likert-scale responses. The analysis involves descriptive statistics, probability calculations, conditional probability, Bayes' theorem, crosstabulation, chi-square tests, correlation analysis, and data visualisation. These methods help identify patterns, show differences in behaviour, and assess whether demographic factors or service features are linked to online shopping frequency.

The results from this analysis help improve understanding of how different groups use online retail platforms and which factors may influence their level of involvement. These findings are useful for researchers studying consumer behaviour and for businesses aiming to improve online services and develop more effective digital retail strategies.

2. Research Question:

2.1 Main Research Question:

The central research question guiding this study is: "What factors influence how often people shop online?" The purpose of this question is to find the main factors linked to differences in online shopping behaviour among respondents. Online shopping has become an important part of modern commerce, but how often people use it varies between individuals. Some respondents shop online weekly or monthly, while others take part in online shopping only sometimes. By looking at demographic factors and perceived importance ratings, this research

question helps identify patterns that may explain why some groups show higher levels of online shopping activity than others in different parts of the population.

2.2 Supporting Research Questions:

To explore the main research question in more detail, three supporting research questions are included.

- Firstly, does age influence how often people shop online? This question looks at whether different age groups show different patterns of using online shopping services.
- Secondly, does gender affect online shopping behaviour? This question examines whether there is a difference in online shopping frequency between male and female respondents.
- Thirdly, which perceived factors - such as security, delivery speed, convenience, website quality, and product range - have the strongest link with frequent online shopping? These factors may help explain or encourage higher levels of online purchasing.

Answering these supporting questions helps provide a wider and clearer understanding of the main research question.

2.3 Importance of the Topic:

This topic is important because online shopping keeps growing in markets around the world, affecting how people buy products and how businesses plan their strategies. Knowing which factors most influence how often people shop online can help organisations improve website design, make delivery services better, and deal with customer concerns about security and convenience. In addition, studying these patterns helps build a broader understanding of consumer behaviour in online and digital environments.

3. Methodology:

The methodology used in this study follows several clear and organised steps to prepare, process, and analyse the dataset. The dataset used is an online shopping survey with 95 responses, provided in Excel format. It contains demographic details such as age and gender, behavioural variables like online shopping frequency and shopping preference, and several Likert-scale questions that measure the perceived importance of different factors, including website quality, security and privacy, product range, convenience, and delivery performance.

3.1 Data Preparation:

The dataset was first imported into a Python environment using the pandas library. An initial check was completed to understand the number of rows, columns, and data types. This process included viewing the first few records, reviewing column names, and checking for missing values. The dataset had no missing values, so the analysis could continue without needing to fill in or replace any data. (McKinney, 2010)

To make the dataset easier to read and analyse, several column names were simplified. Long survey question labels were changed to shorter and more consistent names. Likert-scale responses showing importance levels (for example, "Extremely important" or "Slightly important") were converted into numerical values on a scale from 1 to 5. This change allowed the use of numerical summaries, correlation analysis, and statistical tests.

Extra variables were also created to support deeper analysis. Age groups were converted into approximate numerical midpoints. Online shopping frequency categories were changed into numerical scores from 0 (not at all) to 3 (weekly). A binary variable called "frequent shopper" was then created to separate respondents who shop monthly or weekly from those who shop yearly or not at all. Binary variables were also created for gender (female = 1, male = 0) and shopping preference (online = 1, in-store = 0).

3.2 Analytical Techniques:

Several analytical techniques from the module were used in this study. Descriptive statistics, such as frequency tables and proportions, were used to give an overview of demographic characteristics and online shopping behaviour patterns. Bar charts were created to show gender distribution, age group representation, shopping frequency levels, and preferences for online versus in-store shopping.

3.2.1 Bayes' Theorem:

Probability concepts were applied to calculate empirical probabilities for variables including gender, age groups, shopping frequency, and shopping preference. Conditional probability was used to find values such as the probability of being a frequent shopper given a specific gender. Bayes' theorem was also applied to show how conditional probabilities are related in this situation. (Rossi, Valeri & Baggio, 2022)

3.2.2 Chi-Square:

Crosstabulations were produced to explore relationships between categorical variables, such as gender and frequent shopping status. These tables were then used to carry out chi-square statistical tests (Virtanen *et al*., 2020). A chi-square test for independence was performed to examine whether gender and shopping preference were linked to differences in online shopping frequency. In addition, a chi-square goodness-of-fit test was used to check whether the distribution of shopping frequency followed a uniform pattern or showed clear imbalance. (Sarulatha & Sasirekha, 2024)

3.3 Visualisation Techniques:

Visualisation techniques were used to help interpret both categorical and numerical patterns in the dataset. Bar charts were created to show key distributions, including gender, age groups, shopping frequency categories, and shopping preferences. These visuals made it easier to see differences within the population that may not be clear from numbers alone. A horizontal bar chart was also used to show the mean importance ratings of perceived factors, allowing clear comparison between security, delivery, convenience, website quality, and product range. In addition, a heatmap was produced to display the correlation matrix, highlighting the strength and direction of relationships between importance scores and shopping frequency. Overall, these visual tools improved understanding of the dataset by turning statistical results into clear and easy-to-read graphs. (Ramchandani & Chhabra, 2025)

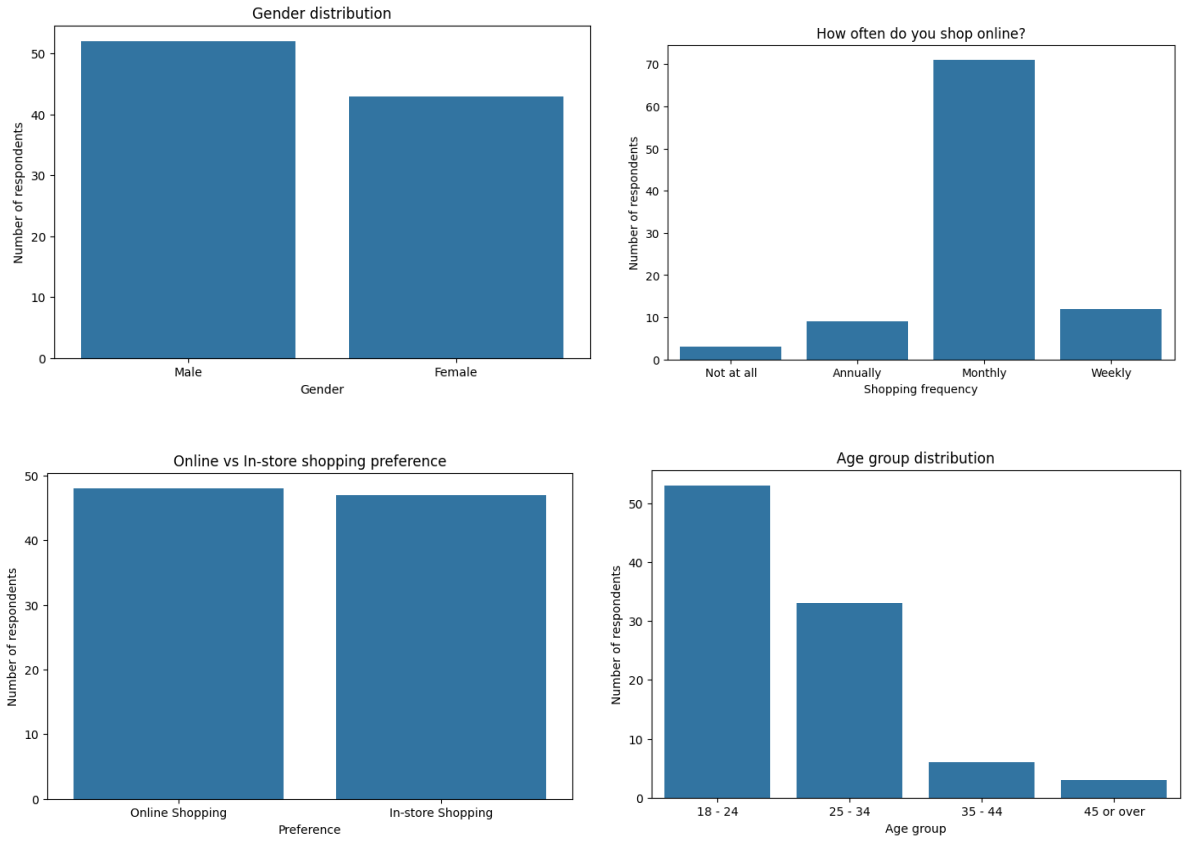
4. Analysis and Presentation of Findings:

4.1 Interpretation of Descriptive Results:

The descriptive analysis showed several clear patterns in the respondent group and their online shopping behaviour. The gender distribution showed a small imbalance, with slightly more male respondents than female respondents. The age distribution showed a strong concentration of respondents aged 18–24, while older age groups appeared in much smaller numbers. This uneven structure suggests that the behaviour patterns seen in the dataset may be strongly influenced by younger users.

Online shopping frequency followed a clear trend, with monthly shopping being the most common category. Weekly shopping occurred less often, while yearly shopping and “not at all” responses made up only a small part of the sample. This pattern suggests that most respondents use online shopping regularly, but not very frequently. The shopping preference variable, which

compared online and in-store shopping, showed only a small difference, indicating a balanced overall preference within the sample.



4.2 Interpretation of Probability and Conditional Probability Findings:

The empirical probability results gave useful insight into how likely certain shopping behaviours were. The probability of a respondent shopping monthly was much higher than the probability of shopping yearly or weekly. In a similar way, the probability of preferring online shopping was slightly higher than preferring in-store shopping, but the difference between the two was small.

```
Frequent shopper counts:
frequent_shopper
1    83
0    12
Name: count, dtype: int64

Frequent shopper proportions:
frequent_shopper
1    0.873684
0    0.126316
Name: proportion, dtype: float64

Gender proportions:
gender
Male      0.547368
Female    0.452632
Name: proportion, dtype: float64

Preference proportions:
shopping_preference
Online Shopping      0.505263
In-store Shopping    0.494737
Name: proportion, dtype: float64
```

Conditional probability was used to check whether demographic factors affected shopping frequency. For example, the conditional probability of being a frequent shopper based on gender showed very similar values for male and female respondents. This suggests that gender has little effect on how often people shop online, even though the total numbers for each group are slightly different. These findings were also supported by Bayesian calculations, which confirmed that gender does not greatly change the likelihood of frequent online shopping.

```
Gender vs frequent shopper table:
 frequent_shopper  0   1
gender
Female            7  36
Male              5  47

P(Female) = 0.453
P(Frequent shopper) = 0.874
P(Frequent | Female) = 0.837
P(Female | Frequent) = 0.434
```

4.3 Interpretation of Chi-Square Test Results:

The chi-square test for independence was used to check whether gender and frequent shopping status were statistically linked. The test gave a very small chi-square value and a p-value higher than the standard 0.05 significance level, showing no meaningful relationship between the two variables. This means that differences in frequent shopping behaviour between male and female respondents are likely caused by random sampling rather than a real difference. As a result, gender does not appear to be an important factor in how often people shop online. (Virtanen *et al*., 2020)

```
Gender vs frequent shopper contingency table:
 frequent_shopper  0   1
gender
Female            7  36
Male              5  47

Expected counts under independence:
 frequent_shopper         0          1
gender
Female            5.431579  37.568421
Male              6.568421  45.431579

Chi-square statistic: 0.439
Degrees of freedom: 1
P-value: 0.507
```

A second chi-square independence test investigated whether shopping preference (online versus in-store) was connected to frequent shopping behaviour. Unlike the gender test, this analysis produced a statistically significant chi-square value with a p-value below 0.05. This

indicates that shopping preference is not evenly distributed across frequent and non-frequent shoppers. Respondents who prefer online shopping are more likely to belong to the frequent shopper group, whereas those who prefer in-store shopping are less likely to shop frequently. This suggests that shopping preference may be an influential factor in determining online shopping habits.

```
Preference vs frequent shopper contingency table:
 frequent_shopper      0   1
shopping_preference
In-store Shopping    10  37
Online Shopping       2  46

Expected counts under independence:
 frequent_shopper            0          1
shopping_preference
In-store Shopping    5.936842  41.063158
Online Shopping      6.063158  41.936842

Chi-square statistic: 4.844
Degrees of freedom: 1
P-value: 0.028
```

A chi-square goodness-of-fit test was used to check whether the four shopping frequency categories followed an even pattern. The results showed a very uneven distribution, with monthly shopping happening much more often than the other categories. The test produced a very large chi-square value and a very small p-value, confirming that the observed pattern is far from what would be expected if all categories were equally likely. This supports the descriptive results and shows a clear behaviour trend in the dataset, where respondents strongly prefer monthly online shopping.

```
Observed counts:
 shopping_frequency
Not at all     3
Annually       9
Monthly       71
Weekly        12
Name: count, dtype: int64

Expected counts under uniform distribution: [23.75 23.75 23.75 23.75]

Chi-square GOF statistic: 127.105
P-value: 2.274e-27
```

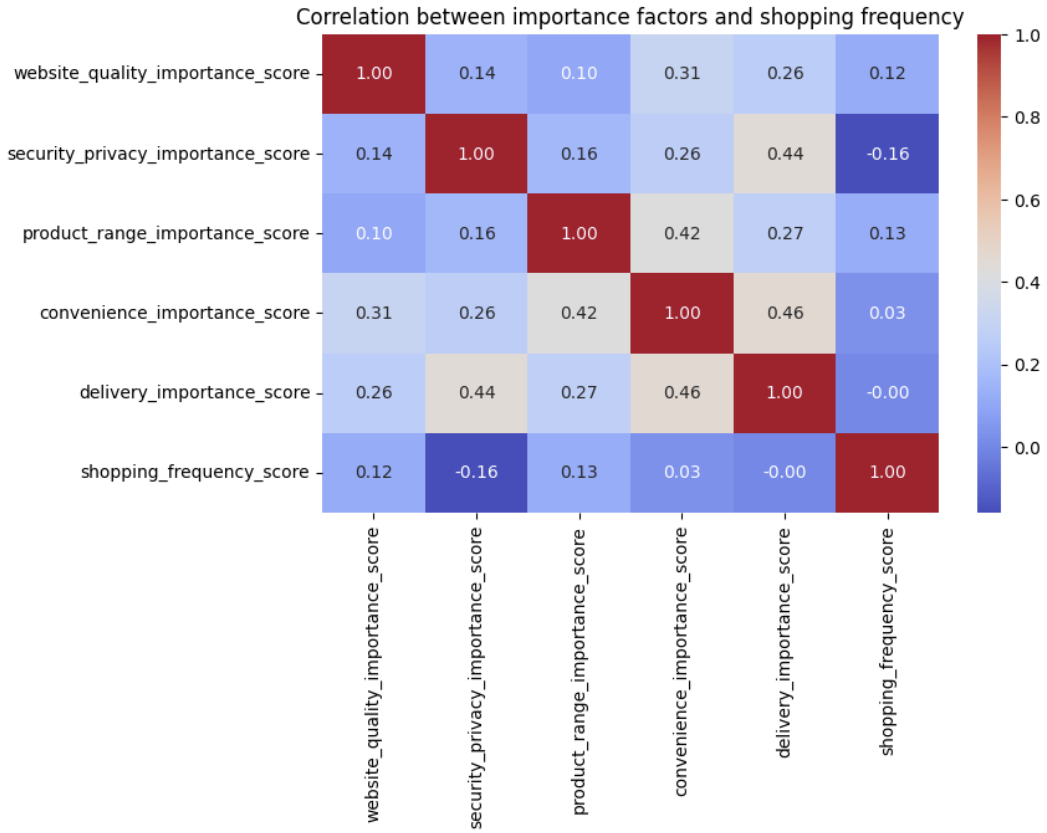
4.4 Correlation Patterns:

Correlation analysis was carried out using numerical values taken from the Likert-scale importance scores. The correlation matrix showed weak positive relationships between perceived service factors and online shopping frequency. Although factors such as website quality, convenience, and product range received high importance ratings, these scores had limited ability to explain how often people shop online. This suggests that frequent online

shopping is more likely influenced by habits or specific situations rather than by how people rate service features.

The strongest correlations were found between the importance ratings themselves. This means that respondents who rated one factor as important usually rated other factors as important as well. While this shows consistent attitudes toward online shopping conditions, it does not appear to lead to higher levels of online shopping frequency.

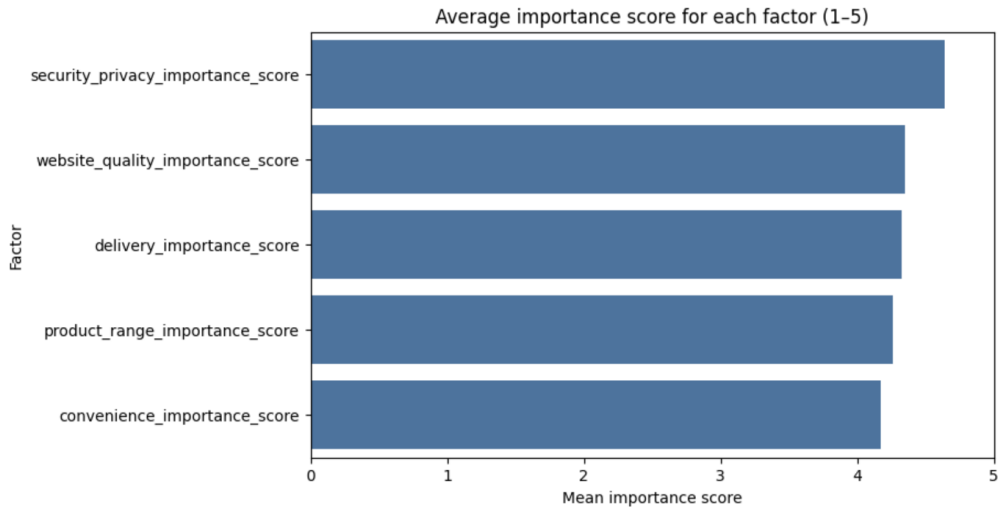
```
Correlation with shopping_frequency_score:

website_quality_importance_score     0.118106
security_privacy_importance_score   -0.159422
product_range_importance_score       0.125871
convenience_importance_score         0.028411
delivery_importance_score           -0.000381
shopping_frequency_score             1.000000
Name: shopping_frequency_score, dtype: float64
```



The average importance scores for online shopping factors show that respondents rated all factors highly, with mean values above four on a five-point scale. Security and privacy received the highest score, showing that safe payments and data protection are very important to online shoppers. Website quality, delivery reliability, and product range were also rated highly, which suggests that users value easy navigation, reliable delivery, and a wide choice of products. Convenience, although still rated positively, had the lowest average score among the five

factors. Overall, these results show that respondents consider many parts of the online shopping process important, with strong focus on trust and ease of use.

```
security_privacy_importance_score    4.642105
website_quality_importance_score     4.347368
delivery_importance_score            4.326316
product_range_importance_score       4.263158
convenience_importance_score         4.168421
dtype: float64
```



4.5 Key Insights:

Several important insights can be taken from the combined results. First, the dataset shows that online shopping frequency is higher among younger respondents. This matches common industry trends, as younger people are usually more active on digital platforms. Second, gender does not seem to have a strong effect on how often people shop online, even though small numerical differences are present. Third, the perceived importance of online shopping factors is generally high but shows little connection with actual shopping behaviour. This suggests that factors such as convenience, website quality, and security are seen as basic expectations rather than key reasons for shopping online more often.

4.6 Limitations of the Dataset:

The dataset has several limitations. The small sample size of 95 respondents limits how widely the results can be applied. The age distribution is strongly focused on younger people, which may reduce how representative the findings are. In addition, the data comes from self-reported survey answers, which can include bias, such as poor memory or giving socially acceptable responses. The uneven spread of shopping frequency categories also reduces the reliability of statistical comparisons between different groups.

4.7 Limitations of Methods:

The statistical methods used in this study are suitable for categorical data, but they also have some limits. Chi-square tests are sensitive to small, expected values, which can lower the reliability of results when group sizes are small. Correlation analysis based on Likert-scale data may simplify complex behaviour patterns too much. In addition, probability calculations rely fully on the observed data, so the results may reflect patterns specific to this sample rather than wider population trends.

4.8 Visual Presentation and Clarity of Results:

The visualisations used in this study greatly improved how easy the data was to understand. Bar charts clearly showed demographic patterns and shopping behaviour distributions, while the heatmap displayed the correlation structure in a simple and clear way. These visuals supported the written explanations and helped make the findings easier to follow. Showing results in graphical form also made it easier to spot patterns, such as the age imbalance in the sample and the dominance of certain shopping frequency categories.

5. Future Work:

Future work in this area could build on the current analysis by dealing with several limitations and widening the scope of the study. One main priority for future research is to increase the sample size so that the results are more statistically reliable. The current dataset includes 95 responses, with a much higher number of younger participants. Collecting a larger and more balanced sample would allow more accurate comparison between demographic groups and reduce the effect of sampling bias. Expanding the dataset to cover wider geographic areas, different age groups, and varied socioeconomic backgrounds would improve how representative the data is and support more general conclusions.

Another important direction for future research is the inclusion of additional explanatory variables. Adding information on income, education level, employment status, device usage, and access to digital technology could help reveal more detailed relationships between personal characteristics and online shopping frequency. Behavioural data, such as real purchase records or browsing behaviour, could also provide more objective insight and help reduce the limitations linked to self-reported survey responses.

Future studies may also apply more advanced analytical methods when larger datasets are available. Techniques such as logistic regression, decision trees, or clustering analysis could help identify deeper behaviour groups and uncover more detailed patterns in online shopping

behaviour. These approaches may improve predictions of frequent shopping activity and provide stronger insight into the factors that drive user engagement.

6. Conclusion:

The analysis of the online shopping survey dataset gives a clear picture of respondent demographics, shopping behaviour, and perceived factors that may affect how often people shop online. The descriptive results showed that most respondents were in the 18–24 age group, meaning younger consumers were strongly represented in the sample. Online shopping frequency was mainly monthly, which suggests that most respondents use online platforms regularly but not very often. Shopping preference was balanced, with only a small tendency toward online shopping compared to in-store buying.

Probability calculations demonstrated that frequent shopping behaviour is more common than occasional shopping within this respondent group. Conditional probability and Bayesian results further indicated that gender plays a minimal role in determining shopping frequency, as both male and female respondents showed similar likelihoods of being frequent shoppers. This finding was reinforced by the chi-square test for independence, which confirmed that no statistically significant association exists between gender and online shopping frequency.

Analysis of perceived importance factors showed consistently high ratings for convenience, security, delivery, product range, and website quality. However, correlation results revealed only weak relationships between these perceived factors and shopping frequency. This suggests that while respondents value these elements, they do not strongly influence how often individuals shop online. Instead, habitual behaviour or situational needs may play a more substantial role.

Taken together, the results show that age distribution and behavioural patterns are more distinct indicators of online shopping frequency than demographic characteristics such as gender or perceived service attributes. The findings highlight the importance of considering user habits and demographic composition when interpreting online shopping trends. Although the dataset provides valuable insights, further research with a larger and more diverse sample would enhance the generalisability and depth of understanding regarding consumer engagement with online retail platforms.

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